TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Digital twin and model-based state estimation for high-efficiency, long-lifetime vanadium redox flow battery energy storage systems (VRFB-ESS)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics (전기·전자)		Electrical & Electronics (전기·전자)	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries (전지)	Secondary Batteries (이차전지)	Heavy Electrical Equipment (중전기기)	Energy storage equipment (에너지저장기기)
Concept of Proposed Technology	Proposed core technology (TNO): a digital-twin-driven monitoring, diagnostics and optimisation stack for VRFB-ESS, enabling higher round-trip efficiency, longer lifetime, and reduced operational risk through physics-informed state estimation and anomaly detection. Key technical elements: 1) VRFB multi-physics models: electrochemistry (Nernst potentials, kinetics, ohmic and mass-			

	transport losses), species crossover and electrolyte imbalance, thermal and hydraulic (flow distribution, pump parasitic power) subsystems. Reduced-order model variants will be created for embedded and cloud execution [3][5]. 2) State-of-X (SoX) estimation for flow batteries: state of charge per electrolyte tank, stack internal resistance and overpotential split, crossover/imbalance indicators, pump and auxiliary losses, and remaining useful life. Algorithms will combine physics-based models with data-driven adaptation (for example, parameter tracking and self-adaptive learning) [3][6]. 3) Model-based anomaly detection: early warning for abnormal degradation and operational faults (for example, shunt currents, imbalance growth, sensor drift, pump failures, unusual efficiency losses) using TNO's model-based detection approaches developed in battery diagnostics and battery-passport demonstrators [2][6]. 4) Validation workflow: test and validation protocols combining laboratory characterisation and hardware-in-the-loop (HIL) with real operational data to identify model parameters and validate estimators and controllers [1][3]. 5) Data and compliance layer: secure, standard-ready data outputs for EU Battery Passport and related digital product passport requirements, including links between battery state algorithms and passport data models [2][13][14][15]. Application and use: - Integration into the Korean SME partner's VRFB energy storage system controller (BMS or EMS) as a software toolchain and demonstrator, enabling operational optimisation (efficiency and lifetime) and predictive maintenance for stationary ESS fleets.
Rationale for Joint R&D Project	Why this joint R&D is needed and why collaboration is required: - VRFBs are among the most technically mature flow-battery options for long-duration stationary storage [7][8]. Reported round-trip efficiencies for VRFB and flow-battery systems vary strongly with operating point and auxiliary loads (for example, pumping and power conversion), making system-level optimisation and real-time diagnostics central to cost and performance [9][12]. - DOE field performance reporting highlights a goal of round-trip plant efficiencies of 80% or better for large VRFB deployments and discusses operational factors influencing achievable efficiency [9]. Achieving and sustaining such performance in real operation requires advanced monitoring and control beyond standard voltage-current supervision. - NREL has shown that VRFB system efficiency is a nonlinear function of active power and state of charge, and is impacted by chemical losses, auxiliary parasitics, and power conditioning losses. This motivates a dynamic system model and control strategy [12]. - The EU Battery Regulation introduces a mandatory battery passport for industrial batteries (including stationary storage) from 18 February 2027, increasing the need for secure, standardised, high-quality operational data and state information [13][14][15].

	Why TNO: - TNO combines battery modelling and state estimation algorithms with test and validation capabilities. TNO's battery portfolio includes BMS-suitable physics-based models, SoX algorithm development (SOC, SOH, SOE, RUL, state of safety), active diagnostics (including EIS approaches), and hardware-in-the-loop and power testing facilities [1][3]. - TNO has active work on battery passport demonstrators that connect battery state algorithms with secure data exchange frameworks [2], and broader expertise in reliable and real-time digital twin systems and lifecycle management [5]. Why the Korean SME partner: - The partner provides a VRFB product and operational context, system data, and pilot or demonstration hardware needed to validate and de-risk the digital twin and algorithms in a relevant environment.
Key Words	1. Vanadium redox flow battery energy storage systems (VRFB-ESS) 2. Digital Twin 3. State estimation 4. Diagnostics/Anomaly Detection 5. Battery Passport

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)
R&D Objectives	R&D targets (technical, measurable): 1) Digital twin model set for VRFB-ESS: validated stack plus balance-of-plant model and reduced-order real-time model, with documented accuracy against measured data (voltage, current, electrolyte temperature, flow rate, pump power) [12]. 2) Real-time SoX estimator suite for VRFB: tank SOC estimation error target < 2% absolute SOC under representative operating profiles (definition and test protocol to be agreed), and detection of electrolyte imbalance or crossover trends with actionable early-warning thresholds. 3) Efficiency optimisation: demonstrate an operational strategy that can achieve and sustain round-trip efficiency targets aligned with state-of-the-art expectations for VRFB plants (>= 80% at representative operating points, subject to system configuration), by reducing auxiliary losses through pump and control optimisation [9][12]. 4) Battery passport data readiness: produce a secure data schema and interface description mapping VRFB operational and state variables to battery passport style requirements and data exchange practices [2][13]. Success criteria will be verified via jointly agreed validation test plans, including HIL and field data evaluation [1][3].			
R&D Target Specifications	R&D details (work packages): WP1 Requirements and baseline (Months 1-3): VRFB system architecture, sensor set, control interfaces, and operating profiles. Define validation protocols and KPIs (efficiency, availability, diagnostics latency). WP2 Model development (Months 1-12): develop VRFB multi-physics model and reduced-order variants. Include electrochemical, thermal, and hydraulic subsystems and identify key parameters for online tracking [12]. WP3 SoX estimation and diagnostics (Months 4-18): develop SOC, SOH and imbalance estimators; implement model-based anomaly detection and sensor validation; integrate data-driven adaptation methods where beneficial [2][6]. WP4 Optimisation and control (Months 10-24): implement optimisation strategies (for example, model predictive control, pump scheduling) to reduce parasitic losses and maintain efficiency across power and SOC ranges [12]. WP5 Integration and validation (Months 16-30): integrate toolchain with partner controller; execute HIL and system tests; calibrate models using lab and field data; validate reliability and performance [1][3]. WP6 Demonstrator and exploitation plan (Months 28-36): demonstrator deployment in			

	operational environment; final validation report; roadmap for productisation, including compliance and data governance [13][14].
R&D Plan	 R&D strategy plan and milestones: M1 (Month 3): System requirements and KPI definition complete; data access and interface specification agreed. M2 (Month 12): VRFB multi-physics model and reduced-order real-time model validated on lab or recorded operational datasets. M3 (Month 18): SoX estimation and anomaly detection algorithms validated in offline replay and HIL environment. M4 (Month 24): Control and optimisation strategies implemented and benchmarked; preliminary efficiency improvement demonstrated. M5 (Month 30): Integrated prototype toolchain running on partner controller or edge platform with live data; field-trial readiness review. M6 (Month 36): Operational demonstrator completed; final performance validation; exploitation and standardisation plan delivered. Project management approach: joint technical steering meetings (monthly), shared model and data repositories, and staged validation gates to manage risk and ensure reproducibility.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities: - Lead development of VRFB digital twin models (multi-physics and reduced-order). - Develop SoX estimation and model-based diagnostics algorithms and provide software implementations. - Design validation protocols and execute HIL and laboratory validation where applicable using TNO Battery Lab methodologies (characterisation, safety markers, HIL testing) [1][3]. - Develop battery passport aligned data model and secure data exchange interface description (in cooperation with SME partner) [2][13]. - Coordinate technical work packages and dissemination (public deliverables, subject to IPR).
Expected R&R of Korean SME Partner	Korean SME partner responsibilities: - Provide VRFB system architecture, operating profiles, sensor signals and historical operational data for model identification and validation. - Provide access to VRFB stack or system for integration and testing, including controller platform and relevant interfaces (BMS or EMS). - Support definition of use cases and commercial requirements (grid services, long-duration storage operation). - Co-execute field validation and support deployment of demonstrator in operational environment. - Contribute to exploitation plan and productisation pathway, including market and regulatory requirements relevant to export.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: - Validated VRFB digital twin and reduced-order real-time model library, including documentation and parameter identification workflow. - Real-time SoX estimation and anomaly detection suite tailored to flow batteries, with implementation guidance for embedding or cloud deployment. - Integrated prototype demonstrator connected to partner VRFB controller, enabling efficiency and lifetime optimisation and predictive maintenance. - Battery passport style data schema and secure interface specification, supporting EU battery passport compliance readiness [2][13][14][15]. Impact (technological, commercial, industrial): - Higher achievable and sustained round-trip efficiency and reduced auxiliary energy consumption through model-based optimisation (addressing documented dynamic efficiency drivers in VRFB systems) [12]. - Improved availability and lower OPEX through early detection of faults and abnormal degradation, enabling condition-based maintenance and reducing unplanned downtime. - Strengthened export readiness to the EU by aligning with battery passport requirements and data governance expectations [13][14][15]. - Transferable methods applicable to other stationary energy storage technologies and next-generation flow-battery chemistries.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description
·A consistent and objective indicator of the technological maturity of key element technologies

	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization